

DevOps Training Program

Azure VM Setup Guide

Account Creation → Virtual Machine → Java → Jenkins Installation

Companion guide to the AWS EC2 + Jenkins setup

Follow this BEFORE attempting the Azure version of your weekly labs

What You'll Do in This Guide

Section	Description
Part 1	Create a Free Azure Account
Part 2	Create a Resource Group
Part 3	Create & Configure a Linux Virtual Machine
Part 4	Connect to the VM via SSH
Part 5	Configure the Network Security Group (Firewall)
Part 6	Install Java (JDK) on the VM
Part 7	Install & Start Jenkins
Part 8	Access Jenkins from Your Browser

REQUIRES	A valid email address, phone number for verification, and a credit/debit card (Azure Free Tier requires this for identity verification — you will NOT be charged unless you exceed free limits or manually upgrade).
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PART 1

Create a Free Azure Account

1. Go to <https://azure.microsoft.com/free> in your browser.
2. Click "Start free" or "Try Azure for free."
3. Sign in with an existing Microsoft account, or click "Create one!" to make a new Microsoft account using your email.
4. Fill in your identity verification details: phone number (OTP verification) and a credit/debit card.
5. Agree to the subscription terms and click "Sign up."
6. Wait for account provisioning — you'll land on the Azure Portal at <https://portal.azure.com>

FREE TIER

New accounts get \$200 credit for 30 days plus 12 months of select free services. A B1s VM (1 vCPU, 1GB RAM) is typically covered under free tier limits — perfect for these labs.

WARNING

Always set up a Budget Alert (Cost Management + Billing → Budgets) right after signup to avoid surprise charges once your free credit expires.

PART 2

Create a Resource Group

A Resource Group is a logical container for all your Azure resources (VM, network, disk, etc.). Creating one first keeps things organized and makes cleanup easy at the end of the course.

7. In the Azure Portal search bar (top), type "Resource groups" and select it.
8. Click "+ Create."
9. Subscription: Azure subscription 1 (default free trial subscription)
10. Resource group name: devops-lab-rg
11. Region: Choose the region closest to you (e.g., East US, Central India, West Europe)
12. Click "Review + create" → "Create"

TIP

Keep ALL your lab resources (VM, disk, network, public IP) inside this same Resource Group. At the end of the course, deleting this one Resource Group removes everything and stops all billing instantly.

PART 3

Create & Configure a Linux Virtual Machine

Step 1 — Start VM Creation

13. In the Azure Portal search bar, type "Virtual machines" and select it.
14. Click "+ Create" → "Azure virtual machine."

Step 2 — Basics Tab

- Subscription: Azure subscription 1
- Resource group: devops-lab-rg (created in Part 2)
- Virtual machine name: devops-jenkins-vm
- Region: Same as your resource group
- Availability options: No infrastructure redundancy required
- Image: Ubuntu Server 22.04 LTS - x64 Gen2
- Size: Click "See all sizes" → select Standard_B1s (1 vCPU, 1 GiB RAM — Free tier eligible)

Step 3 — Administrator Account

- Authentication type: SSH public key (recommended) — or Password for simplicity if preferred
- Username: azureuser
- SSH public key source: Generate new key pair
- Key pair name: devops-vm-key

When you click "Review + create" later, Azure will prompt you to Download private key and create resource — SAVE this .pem file safely. You cannot download it again.

Step 4 — Inbound Port Rules

- Public inbound ports: Allow selected ports
- Select inbound ports: SSH (22) and HTTP (80)

(We will add port 8080 for Jenkins separately in Part 5, for stricter access control practice.)

Step 5 — Disks & Networking Tabs

- Disks tab: Leave default OS disk type (Standard SSD) — click Next
- Networking tab: Leave default Virtual Network and Subnet — confirm Public IP is set to "New" — click Next

Step 6 — Review and Create

15. Click "Review + create." Wait for validation to pass (green checkmark).
16. Click "Create."

17. When prompted, click "Download private key and create resource" — save devops-vm-key.pem.
18. Wait 1–2 minutes for deployment to complete, then click "Go to resource."
19. Note down the Public IP address shown on the VM Overview page.

PART 4

Connect to the VM via SSH

On Mac/Linux/WSL terminal, secure the key file permissions first:

```
chmod 400 devops-vm-key.pem
```

Connect via SSH using the public IP from Part 3:

```
ssh -i devops-vm-key.pem azureuser@<VM-Public-IP>
```

On Windows (PowerShell), if chmod isn't available, right-click the .pem file → Properties → Security → restrict access to your user only, then run the same ssh command.

Once connected, verify the system:

```
whoami && uname -a  
sudo apt update -y
```

NOTE

Unlike AWS's ec2-user or ec2-user@amazon, Azure's default Ubuntu image uses the username you set during VM creation (azureuser in this guide) and apt instead of yum, since Ubuntu is Debian-based.

PART 5

Configure the Network Security Group (Firewall)

Azure's equivalent of an AWS Security Group is a Network Security Group (NSG). We need to open port 8080 for Jenkins.

20. In the Azure Portal, go to your VM (devops-jenkins-vm) → left menu → Networking.
21. Click "Add inbound port rule."
22. Source: Any (or My IP for tighter security) | Destination port ranges: 8080
23. Protocol: TCP | Action: Allow
24. Priority: 310 (or any free number) | Name: Allow-Jenkins-8080
25. Click "Add."

TIP

This step is the Azure equivalent of adding a custom TCP rule for port 8080 in an AWS Security Group — same concept, different portal.

PART 6

Install Java (JDK) on the VM

Jenkins requires Java to run. Jenkins (recent versions) requires Java 17 or 21.

```
sudo apt update -y  
sudo apt install -y fontconfig openjdk-17-jre
```

Verify the installation:

```
java -version
```

Expected output should show something like: openjdk version "17.0.x"

PART 7

Install & Start Jenkins

Step 1 — Add the Jenkins Repository

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
"https://pkg.jenkins.io/debian-stable binary/" | \
sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

Step 2 — Install Jenkins

```
sudo apt update -y
sudo apt install -y jenkins
```

Step 3 — Start & Enable Jenkins

```
sudo systemctl start jenkins
sudo systemctl enable jenkins
sudo systemctl status jenkins
```

Confirm the status shows "active (running)" in green.

Step 4 — Retrieve the Initial Admin Password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Copy this password — you'll need it in the next step.

PART 8

Access Jenkins from Your Browser

26. Open a browser and navigate to: `http://<VM-Public-IP>:8080`
27. Paste the Initial Admin Password copied from Part 7, Step 4.
28. Click "Install suggested plugins" and wait for setup to complete.
29. Create your first Admin User (username, password, name, email).
30. Confirm the Jenkins URL (default is fine) → click "Save and Finish."
31. Click "Start using Jenkins" — you should land on the Jenkins Dashboard.

SUCCESS

If you see the Jenkins Dashboard, your Azure VM is fully set up with Java and Jenkins — ready to run the same Jenkins labs (Freestyle Jobs, Pipelines, GitHub integration) you completed on AWS EC2!

Quick Reference: AWS EC2 vs Azure VM Terminology

Concept	AWS	Azure
Virtual machine	EC2 Instance	Virtual Machine (VM)
Default Linux user	ec2-user / ubuntu	azureuser (custom, set at creation)
Package manager (Ubuntu)	apt	apt
Firewall	Security Group	Network Security Group (NSG)
Login key file	.pem key pair (Key Pairs)	.pem key pair (SSH public key)
Resource container	VPC / no strict grouping	Resource Group
IAM equivalent	IAM Users/Roles/Policies	Azure AD / Microsoft Entra ID + RBAC
Object storage	S3	Blob Storage